NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

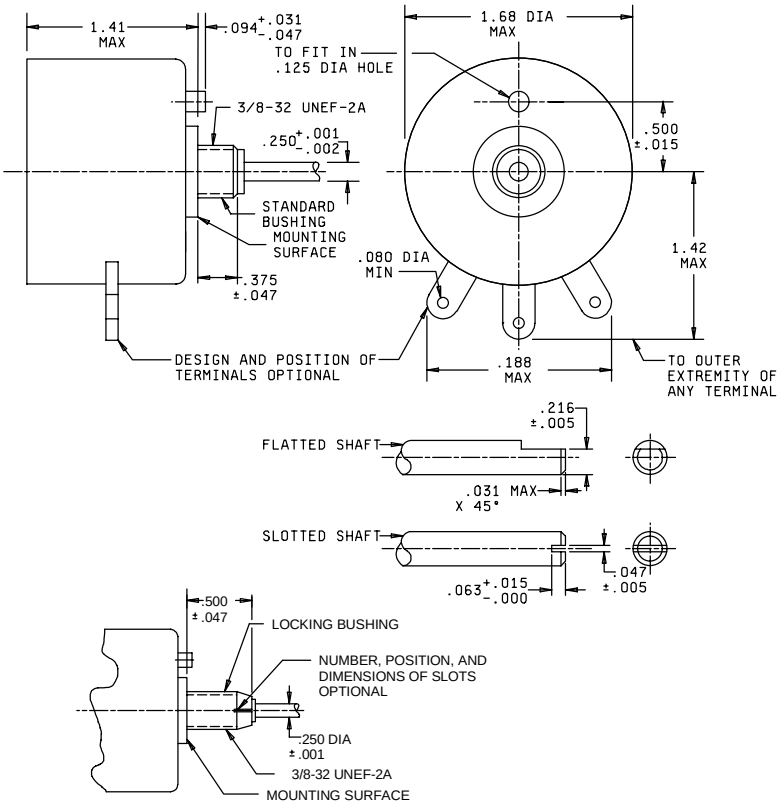
MIL-PRF-22/3B
22 June 1965
SUPERSEDING
MIL-R-22/3A
10 September 1964

PERFORMANCE SPECIFICATION

RESISTOR, VARIABLE, (WIREWOUND, POWER TYPE)

(UNENCLOSED), STYLE RP10

The complete requirements for procuring the resistors described herein shall consist of this document and the latest issue of the Specification MIL-R-22.



INCHES	MM
.001	.03
.002	.05
.005	.13
.015	.38
.031	.79
.047	1.19
.063	1.60
.080	2.03
.094	2.39
.216	5.40
.250	6.35
.375	9.53
.500	12.70
1.410	35.81
1.420	36.07
1.680	42.67
1.880	47.75

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FSC 5905

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TABLE I. Style RP10.

Type designation 1/ (for reference only)	Nominal total resistance value (ohms)	Maximum current (not to be exceeded on any portion of the winding)(amperes)	Part number 1/
RP10---1R0KK	1.0	5.00	M22/03-0025---
RP10---2R0KK	2.0	3.54	M22/03-0001---
RP10---2R5KK	2.5	3.16	M22/03-0002---
RP10---3R0KK	3.0	2.89	M22/03-0003---
RP10---5R0KK	5.0	2.24	M22/03-0004---
RP10---6R0KK	6.0	2.04	M22/03-0005---
RP10---8R0KK	8.0	1.77	M22/03-0006---
RP10---100KK	10	1.58	M22/03-0007---
RP10---150KK	15	1.29	M22/03-0008---
RP10---250KK	25	1.00	M22/03-0009---
RP10---350KK	35	0.85	M22/03-0010---
RP10---500KK	50	0.71	M22/03-0011---
RP10---750KK	75	0.58	M22/03-0012---
RP10---101KK	100	0.50	M22/03-0013---
RP10---151KK	150	0.41	M22/03-0014---
RP10---201KK	200	0.35	M22/03-0015---
RP10---251KK	250	0.32	M22/03-0016---
RP10---351KK	350	0.27	M22/03-0017---
RP10---501KK	500	0.22	M22/03-0018---
RP10---751KK	750	0.18	M22/03-0019---
RP10---102KK	1,000	0.16	M22/03-0020---
RP10---152KK	1,500	0.13	M22/03-0021---
RP10---252KK	2,500	0.10	M22/03-0022---
RP10---352KK	3,500	0.08	M22/03-0023---
RP10---502KK	5,000	0.07	M22/03-0024---

1/ The complete type designation and part number include symbols indicating electrical off position from table II, style of shaft and type of mounting from table III, and length of operating shaft from table IV. (See example of part number.)

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TABLE II. Electrical off position.

Symbol	Electrical off position
1 - - - - - - - - - - - - - -	No electrical off position.
2 - - - - - - - - - - - - - -	Electrical off position at end of rotation of control knob in a counterclockwise direction.
3 - - - - - - - - - - - - - -	Electrical off position at end of rotation of control knob in a clockwise direction.

TABLE III. Style of shaft and type of mounting.

Symbol	Style of shaft	1/4-inch diameter shaft	
		Standard bushing	Locking bushing
F - - - - - - - - -	Flatted	X	---
S - - - - - - - - -	Slotted	X	---
T - - - - - - - - -	Flatted	---	X
U - - - - - - - - -	Slotted	---	X

TABLE IV. Length of operating shaft.

Standard length of shaft measured from mounting surface of resistor.				
	Flatted		Slotted	
Symbol	1/4-inch diameter shaft		1/4-inch diameter shaft	
	Standard bushing	Locking bushing	Standard bushing	Locking bushing
	Inches, ±3/64	Inches, ±3/64	Inches, ±3/64	Inches, ±3/64
A - - -	---	---	1/2	---
B - - -	---	---	---	5/8
D - - -	7/8	---	7/8	---
G - - -	---	1-1/4	---	1-1/4
H - - -	1-1/2	---	1-1/2	---
J - - -	2	---	2	---
K - - -	1/ 2-1/2	---	2-1/2	---
N - - -	4	---	---	---
R - - -	6	---	---	---

1/ This shaft shall be flatted to within 0.156 inch of the mounting bushing.

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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Resistance value: See table I.

Resistance tolerance: ±10 percent.

Electrical off position: All three off positions applicable. See table II.

Shaft style and type of mounting: See table III.

Shaft length: See table IV.

Mechanical rotation: 290° ± 15°.

Hardware: One 3/8-32 UNEF-2B corrosion-resistant, hexagonal mounting nut having a nominal thickness of 3/32 inch and measuring 9/16 inch across the hexagonal flats and one internal-tooth lockwasher in accordance with MS35333-76. Locking nut for locking-bushing type resistors - 0.151 to 0.234 inch thick, - 1/2 inch across the hexagonal flats; thread size - 3/8-32 UNEF-2B.

Dielectric withstanding voltage:
At atmospheric pressure - Method 301 of MIL-STD-202. Test voltage-1,000 volts root mean square (rms).
At reduced pressure - Method 105 of MIL-STD-202. Test voltage - 550 volts.

DC resistance: Total and minimum. Method 303 of MIL-STD-202.
Total resistance within ±10 percent of the nominal total resistance specified in table I.

Minimum resistance shall not exceed 0.2 percent of the nominal total resistance value or 0.2 ohm, whichever is greater.

Torque:
Operating - Not less than 4 ounce-inches (oz-in), nor more than 2.5 pounds-inches (lb-in). Stop - 15 lb-inch. Locking - Shall withstand 20 oz-in without movement.

Low-temperature exposure:
Torque not in excess of 4 lb-in.

Power rating: 25 watts at 25°C. Mounted on a steel panel 12 inches square by 0.063 inches thick. Hot spot not in excess of 340°C.

Load life: Full wattage (see power rating, above) applied 1-1/2 hours on and 1/2 hour off for 1,000 hours. Change in total resistance not to exceed ±5 percent.

Vibration: Method 201 of MIL-STD-202. Change in total resistance not to exceed ±5 percent. Contact-arm not to move more than 2°. Change in resistance between contact-arm terminal and either end terminal not to exceed ±5 percent. No mechanical damage.

Salt spray (corrosion): Method 101 of MIL-STD-202. Test condition letter B (48 hours).

Humidity (steady state): Method 103 of MIL-STD-202. Test condition letter B (96 hours). Change in total resistance not to exceed ±10 percent.

Life (rotational): Method 206 of MIL-STD-202. Test condition letter A (500 cycles) for locking bushing types. Test condition letter C. (5,000 cycles) for standard bushing types. Change in total resistance not to exceed ±5 percent.

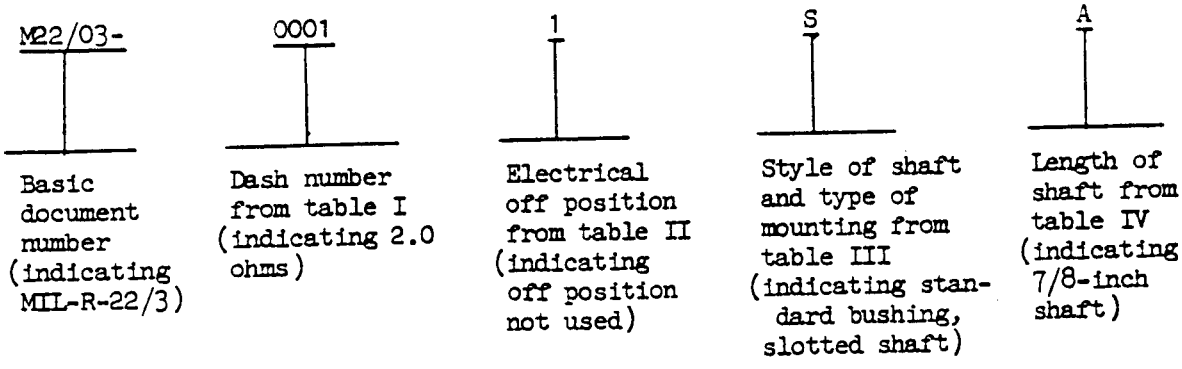
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Terminal strength: Not applicable.

Shock: Method 202 of MIL-STD-202. Change in resistance shall not exceed ±10 percent due to contact arm movement and the total resistance change shall not exceed ±2 percent.

Part number: Consists of the basic number of this specification sheet, a dash number from table I and electrical and mechanical characteristics taken from tables II, III, and IV.

Example:



M22/03- : Basic document number (indicating MIL-R-22/3)

0001 : Dash number from table I (indicating 2.0 ohms)

1 : Electrical off position from table II (indicating off position not used)

S : Style of shaft and type of mounting from table III (indicating standard bushing, slotted shaft)

A : Length of shaft from table IV (indicating 7/8-inch shaft)

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH, WP
Air Force - 11, 17, 85

User activities:
Army - MO
Navy - MC

Preparing activity:
Navy - SH

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